HYPER-LOCAL AVIATION WEATHER FORECASTING AND VERIFICATION SYSTEM FOR SRILANKAN MOONSOON PATTERNS

INDIVIDUAL THESIS

Jayathilaka P.K.V.K.

Student ID: IT22069504

Individual component: QNH, Dew Point and Relative Humidity Prediction Model

Group ID: R26-IT-132

Report submitted in partial fulfilment of the requirements for the bachelor's degree thesis

Sri Lanka Institute of Information Technology

Department of Computing

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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Student name: Jayathilaka P.K.V.K.

Student ID: IT22069504

Signature: _____ Date: 10/10/2026

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Role and name	Signature	Date
Supervisor: Ms.Shashika Lokuliyana		
Co-supervisor: Ms. Hansika Mahaadikara		

ABSTRACT

Reliable short-term aviation weather forecasts are important at Bandaranaike International Airport (VCBI), where rapidly changing tropical and monsoonal conditions can affect operational planning. This individual component documents a three-hour-ahead prediction approach for QNH and dew point, and derives relative humidity (RH) from predicted dew point and predicted dry-bulb temperature using a physical relationship. The wider group system combines airport-focused meteorological data acquisition, forecasting, verification and operational presentation. In the supplied group paper, LightGBM is identified for the QNH and dew-point subsystem. Historical observations are stored in a local SQLite database, ordered chronologically using UTC timestamps, and processed to create time-based and meteorological features. Separate prediction targets are shifted three hours forward, and the train/test split is chronological rather than randomly shuffled. The reported dew-point model achieved an R-squared value of approximately 0.66 and a mean absolute error of approximately 0.58 °C. The QNH model is reported to have an error of approximately 1.8 hPa; the supplied summary does not clearly identify the exact metric represented by that value. RH is calculated from predicted dew point and predicted dry-bulb temperature using the August–Roche–Magnus equation. The subsystem also forms part of an MLOps workflow supporting periodic retraining and live verification against subsequent observations. The thesis documents the problem context, theoretical basis, methodology, design, implementation considerations, available results, limitations and future work. The reported values are treated as results from the group paper rather than as independently reproduced experiments. Further validation across complete monsoon seasons, explicit RH evaluation and reproducible model artefacts are needed before operational use.

Keywords: aviation weather forecasting, QNH, dew point, relative humidity, LightGBM

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my research supervisor for the guidance, constructive feedback and encouragement provided throughout the development of this individual research component. I also acknowledge the support of the wider research group, whose work on the hyper-local aviation weather forecasting and verification system provides the context for this component.

I am grateful to the Sri Lanka Institute of Information Technology and the Faculty of Computing for providing the academic environment in which this research was undertaken. I acknowledge the authors and researchers cited in the group paper for the scientific and technical context that informed this report.

Finally, I acknowledge the importance of accurate documentation, responsible data use and transparent reporting of limitations. Any remaining errors or omissions are my responsibility.

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LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
API	Application Programming Interface
DA	Density Altitude
DL	Deep Learning
hPa	Hectopascal
ICAO	International Civil Aviation Organization
ISA	International Standard Atmosphere
LightGBM	Light Gradient Boosting Machine
MAE	Mean Absolute Error
METAR	Meteorological Aerodrome Report
ML	Machine Learning
MLOps	Machine Learning Operations
QNH	Aviation altimeter setting referenced to mean sea-level pressure
RH	Relative Humidity
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
SQLite	Lightweight relational database engine
TAF	Terminal Aerodrome Forecast
Td	Dew-point temperature
T	Dry-bulb temperature
UTC	Coordinated Universal Time
VCBI	Bandaranaike International Airport ICAO location indicator

1 INTRODUCTION

1.1 Overall Project Context

Aviation weather information is an important input to safe and efficient airport operations. Conditions such as atmospheric pressure, temperature and moisture can change over short periods, particularly in tropical environments affected by monsoonal weather. Reliable observations and well-evaluated short-term forecasts help aviation personnel understand current conditions and anticipate possible changes. However, any prediction system intended for aviation use must present its outputs transparently and must not be treated as a replacement for authorised meteorological services.

This thesis forms part of the group research project, “Hyper-Local Aviation Weather Forecasting and Verification System for Sri Lankan Monsoon Conditions,” developed in the context of Bandaranaike International Airport (VCBI), Sri Lanka. The wider project explores data-driven forecasting and verification for selected weather variables using local meteorological observations. The common system brings together individual research components and provides a shared context for data preparation, forecast generation, evaluation and presentation. The present thesis concentrates on the prediction of QNH, dew point and relative humidity.

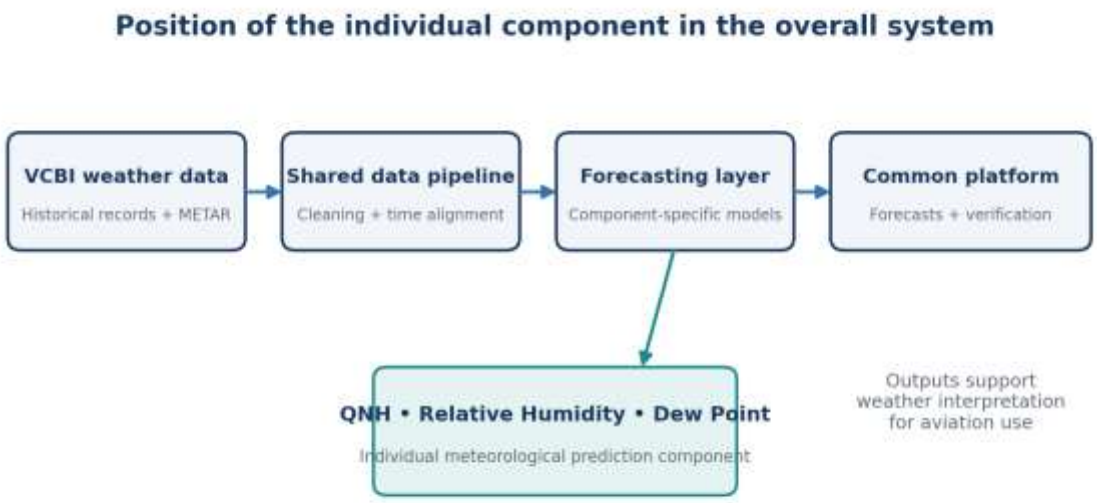


Figure 1-Position of the individual component in the overall system

1.2 Individual Component Context

The individual component investigates short-term prediction of three related meteorological variables: QNH, dew point and relative humidity (RH). QNH is the pressure value adjusted to mean sea-level pressure for aviation altimeter-setting purposes and is commonly expressed in hectopascals (hPa). Dew point is the temperature to which air must be cooled, at approximately constant pressure and water-vapour content, for saturation to occur. Relative humidity expresses the amount of water vapour in the air relative to the saturation amount at the same temperature, as a percentage.

These variables are physically related, but they describe different aspects of the atmosphere. Temperature and dew point are closely connected to moisture conditions, while relative humidity depends on both the actual moisture content and the temperature-dependent saturation vapour pressure. The component therefore needs consistent timestamps, clear units, suitable input features and checks for physically implausible values. The work considers a three-hour forecast horizon in line with the component's stated workflow. The exact model inputs, transformations and outputs should remain consistent with the implementation and experiment records.

1.3 Problem Statement

Historical aerodrome weather observations provide a record of atmospheric conditions, but observations alone do not indicate how the selected variables may change over the following hours. For a hyper-local forecasting system, the challenge is to transform time-stamped historical and recent observations into useful short-term estimates of QNH, dew point and relative humidity while maintaining data quality and traceability. Missing values, inconsistent units, irregular observation intervals, duplicated records and poorly aligned target times can affect model training and the reliability of evaluation.

A further challenge is demonstrating forecast performance using a clearly defined test period and appropriate metrics. Without a reproducible preparation workflow and a comparison between predicted and observed values, a reported result may be difficult to interpret or reproduce. The research problem addressed by this component is therefore how to prepare local weather data and develop and evaluate a three-hour prediction workflow for QNH, dew point and relative humidity, while documenting its limitations and avoiding claims beyond the available evidence.

1.4 Research Gap

The wider project literature establishes the relevance of machine-learning methods and local weather information for short-term forecasting. Nevertheless, results from studies of rainfall, wind, cloud, visibility or general weather forecasting cannot automatically demonstrate performance for QNH, dew point or relative humidity at VCBI. Differences in location, observation frequency, forecast horizon, target definition, preprocessing and evaluation metrics limit direct numerical comparison.

The component addresses this context-specific need by focusing on the three selected variables within a shared hyper-local aviation weather framework and by emphasising data preparation, time alignment, model evaluation and verification. The contribution should be understood as a component-level investigation rather than proof that machine-learning forecasts can replace official aviation weather products. The extent to which the approach improves on a baseline must be established from the final experiment results; it should not be assumed without supporting measurements.

1.5 Aim and Objectives

The aim of this research component is to develop and evaluate a three-hour forecasting workflow for QNH, dew point and relative humidity using local aerodrome weather observations from VCBI, with attention to data quality, reproducibility and forecast verification.

The specific objectives are to:

Prepare and inspect the available meteorological observations, including timestamp consistency, missing values, duplicates, units and plausible value ranges.
Identify and prepare suitable predictors for QNH, dew point and relative humidity in accordance with the implemented feature-engineering pipeline.
Develop or apply the component's selected machine-learning approach for the three-hour prediction task, documenting the training and inference process.
Evaluate the predictions on an appropriate held-out period using relevant metrics and comparisons between predicted and observed values.
Document the component architecture, assumptions, limitations and verification process so that the work can be understood and assessed within the wider system.

1.6 Research Questions or Hypotheses

The component is guided by the following research questions:

RQ1: How can VCBI weather observations be prepared and transformed into consistent inputs for three-hour prediction of QNH, dew point and relative humidity?
RQ2: How well does the implemented forecasting approach predict each target variable on the selected evaluation data, when assessed using appropriate error and goodness-of-fit metrics?
RQ3: What data-quality, temporal-alignment and operational limitations need to be considered when interpreting these component-level forecasts?

Where a formal hypothesis test is not specified in the implementation or experiment plan, this thesis treats these as research questions rather than presenting an unsupported statistical

hypothesis. Any claim that the model outperforms a baseline should be supported by a documented, like-for-like evaluation.

1.7 Scope, Limitations and Assumptions

The scope is limited to the QNH, dew-point and relative-humidity prediction component within the wider VCBI hyper-local aviation weather project. It covers the preparation of relevant historical or recent observations, feature preparation, three-hour model inference, evaluation and documentation of the component's place in the overall system. The work does not claim to provide a certified aviation weather service, replace METAR or TAF products, or make operational decisions on behalf of pilots, air traffic controllers or meteorological officers.

The results depend on the coverage, quality and consistency of the available observations and on the correctness of the implemented preprocessing and modelling steps. Performance may be affected by missing records, changes in observation practices, unusual weather events, limited examples of rare conditions and differences between training and future data. Metrics reported in source materials must be checked against the final experiment outputs, including the target definition, units, sample count and evaluation interval.

The component assumes that timestamps and variable units can be interpreted consistently; that the training and evaluation split prevents future information from leaking into training features; and that predicted values are compared with observations representing the same target time. Relative humidity calculations or derived features must use consistent temperature units and a documented formula. Where these assumptions cannot be confirmed from the available records, they are treated as limitations requiring verification rather than as established facts.

1.8 Individual Contribution

The individual contribution is the QNH, dew-point and relative-humidity prediction component of the group system. It focuses on defining the target variables, preparing the associated meteorological data, supporting the feature and modelling workflow, documenting the logical architecture, and interpreting the component's evaluation results. It also considers how the component's outputs can be stored, verified and presented through the common platform.

The component is intended to integrate with the wider project's shared data and presentation workflow, while maintaining clear separation between its own variables and the other group components, such as wind, temperature/pressure, and cloud/visibility prediction. The thesis reports only the contribution and results that can be supported by the available implementation and experiment evidence. Group-level functionality is described for context and is not claimed as an individual achievement unless it was directly implemented or documented as part of this component.

1.9 Thesis Structure

The remainder of this thesis is organised as follows. Chapter 2 reviews the literature and related work relevant to local aviation weather prediction, the selected meteorological variables and data-driven forecasting. Chapter 3 describes the research methodology, data sources, preparation steps, feature engineering and evaluation measures. Chapter 4 presents the component design and logical architecture. Chapter 5 describes implementation and testing, including the training and inference workflow. Chapter 6 presents and discusses the reported model results and verification considerations. The appendices provide supporting materials such as data definitions, additional evidence, test information and the retraining and verification cycle.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review follows the scope and references of the supplied group paper rather than claiming to be a new systematic review. It focuses on machine learning for weather prediction, aviation forecasting, airport-specific hazards, temporal modelling and the role of observed meteorological data.

The group paper discusses Sri Lankan monsoon rainfall and short-term weather prediction, machine-learning and deep-learning approaches, automated terminal aerodrome forecasting, and aviation weather hazards. These sources provide context but do not all evaluate the same target variables or forecast horizon.

Comparisons across papers must be cautious. Results for rainfall, visibility, wind speed or flight delays cannot be compared numerically with QNH or dew-point errors because geography, target definition, sampling, forecast horizon and metric differ.

For review method and scope, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

2.2 Domain and theoretical foundations

Weather variables describe a coupled physical system. Temperature, moisture, pressure, wind and cloud conditions influence one another, but their relationships are not fixed linear equations across all locations and weather regimes. A data-driven model can estimate patterns from historical observations, while physical equations can provide constraints or derived quantities.

QNH is an aviation altimeter setting referenced to mean sea-level pressure. It should be evaluated in its original unit, normally hectopascals in this project. Correct timestamp alignment is essential because a correct value attached to the wrong hour is still an invalid forecast result.

Dew point is a moisture-related temperature. The difference between dry-bulb temperature and dew point is called dew-point depression. A smaller depression generally indicates air closer to saturation, but fog and visibility also depend on mixing, wind, aerosols and local temperature evolution.

Relative humidity is the ratio of actual water-vapour pressure to saturation vapour pressure at the same temperature, expressed as a percentage. Because saturation vapour pressure varies with temperature, RH depends on both temperature and moisture content. This motivates calculating RH from predicted temperature and dew point.

The project paper uses RH = 100 × exp[(17.67Td/(Td + 243.5)) − (17.67T/(T + 243.5))], where Td is dew point and T is dry-bulb temperature in °C. The implementation must use consistent units and check numerical validity.

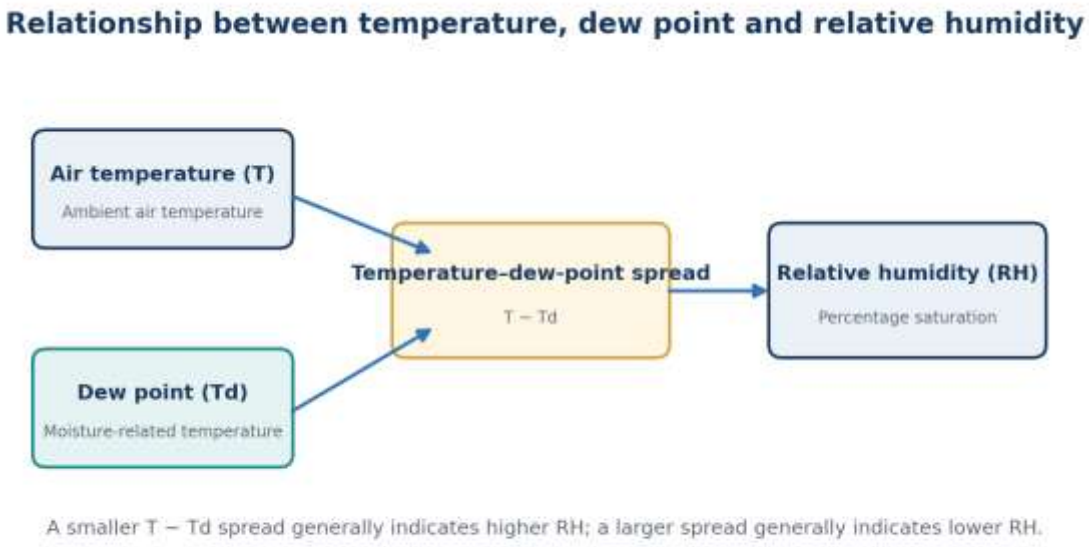


Figure 2-- Relationship between temperature, dew point and relative humidity

For domain and theoretical foundations, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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2.3 Related research

The group paper cites research on monsoon rainfall forecasting in Sri Lanka using artificial neural networks and machine-learning-based short-term weather prediction. These studies support the relevance of data-driven prediction in the national climate context, but their focus is not necessarily the joint airport-level target addressed here.

The paper also discusses ConvLSTM precipitation nowcasting, which is designed to capture spatial and temporal structure in precipitation-related data. The QNH and dew-point component instead uses structured local observations and LightGBM regression. The difference reflects data representation and task rather than a universal ranking of algorithms.

Research on automated terminal aerodrome forecasting illustrates the potential for AI to support aviation forecast generation. Aviation forecasting also has a different risk context from general consumer weather applications, so traceability, verification and communication of limitations are particularly important.

Literature on high-impact aviation weather and airport hazards underlines the need for location-specific weather information. These works do not directly validate this component's QNH or dew-point predictions, but they support the rationale for an airport-focused forecasting system.

For related research, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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2.4 Existing systems and techniques

METAR and TAF serve different roles in the forecasting lifecycle. METAR provides coded observations for an aerodrome at a particular time, while TAF represents a forecast over a defined period. A verification system must distinguish them and match predictions to the appropriate target time.

Numerical weather prediction systems use physical models and data assimilation. Machine-learning models can complement this ecosystem by learning local patterns, post-processing forecasts or generating specialised predictions. This thesis does not establish that LightGBM replaces numerical weather prediction; its purpose is narrower and airport-specific.

Dashboards improve access to observations and forecasts, but they are only as reliable as the underlying data pipeline. Each forecast should carry its issue time, target time, unit, model version and verification status.

For existing systems and techniques, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and

prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

2.5 Gap synthesis and conceptual framework

The conceptual framework begins with time-stamped meteorological observations, followed by data-quality checks, chronological ordering and feature engineering. Separate models estimate QNH and dew point at t+3 hours. A temperature forecast is combined with predicted dew point to calculate RH.

Three boundaries matter. Model predictions are statistical estimates rather than official observations; RH is derived and depends on two forecast inputs; and retraining is a maintenance activity that must be evaluated rather than assumed to improve accuracy.

The contribution is the application and documentation of this linked forecasting pathway for VCBI. The framework supports auditability from input observations through features, model outputs, derived RH and later verification.

For gap synthesis and conceptual framework, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows an applied quantitative design. It develops a software-based forecasting component and evaluates continuous numerical predictions against observed values. The available paper does not provide enough information to claim a fully controlled comparison of every alternative model.

The main modelling choices are target variable, forecast horizon, feature set, algorithm configuration and training interval. MAE, RMSE and R^2 are suitable regression metrics, but should be reported separately by variable because units and meanings differ.

Temporal integrity is the methodological priority. Random shuffling can place future observations in the training set while earlier observations are tested, making performance appear better than real forward forecasting. The paper states that the split is chronological, with 80% training and 20% testing.

For research design, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3.2 Data and study materials

The group paper states that historical VCBI meteorological data from 2019–2026 contain more than 120,000 samples in the wider project. It also describes current METAR

observations obtained from the Aviation Weather Center. The individual component uses a local SQLite database and UTC ordering.

The complete row-level dataset, exact field schema, sampling interval and complete data dictionary are not included in the supplied files. The final report should add the actual database schema, source, collection dates, licensing conditions and any exclusions.

Potential fields include observation timestamp, QNH, dry-bulb temperature and dew point. Additional variables should only be described as implemented if they exist in the actual database and source code.

For data and study materials, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3.3 Experimental procedure

First, load historical observations from SQLite and validate the schema. Parse timestamps, sort chronologically, inspect duplicates, convert variables to numeric types and apply a documented missing-value policy. Impossible values should be flagged for review rather than silently passed into the model.

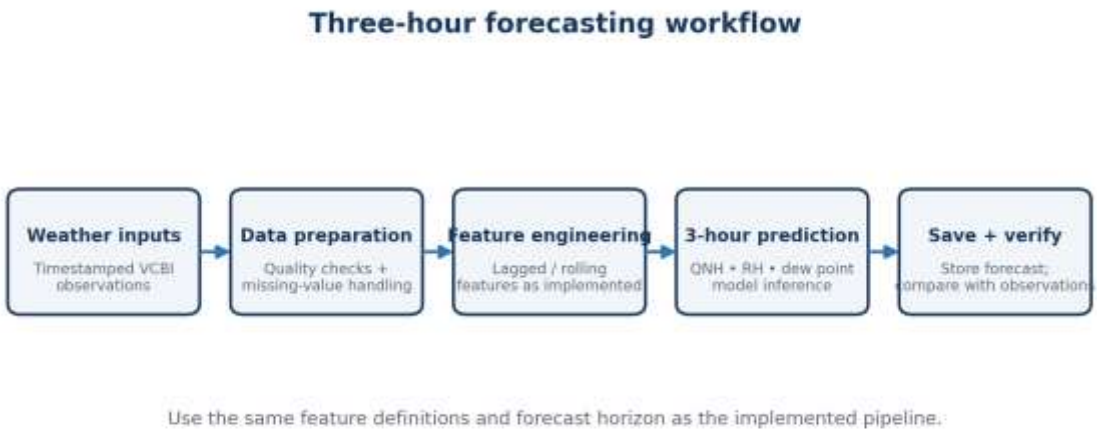


Figure 3 -Three-hour forecasting workflow

Second, engineer time and meteorological features. The paper describes cyclic hour and time-of-year features, temperature-dew-point difference, pressure movement and three-hour and six-hour rolling averages. All features must use only information available at forecast issue time.

Third, create targets at t+3 hours. Timestamp-based matching is safer than a simple row shift if observations are missing or irregular. Exclude examples whose target observation is absent and record how many were excluded.

Fourth, split the data chronologically. Model selection and hyperparameter tuning should use the training interval or a time-aware validation split, not the final test period. The final test interval should remain unseen until the model configuration is fixed.

Fifth, fit separate LightGBM regressors for QNH and dew point. Save the model with its feature list, training period, configuration and software environment. Inference should apply the same feature transformations as training.

Finally, save predictions with issue and target timestamps. When the target observation becomes available, match it to the forecast, calculate errors and update performance records without overwriting the original forecast.

For experimental procedure, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3.4 Evaluation metrics

Mean Absolute Error is the average absolute difference between prediction and observation. It is interpretable in the original unit: °C for dew point and hPa for QNH. It treats each absolute error linearly and is less sensitive to outliers than squared-error metrics.

Root Mean Squared Error is the square root of the mean squared error. It penalises large misses more heavily than MAE and is useful when occasional large errors matter. Both should be reported to give complementary information.

R^2 compares squared prediction error with the variation around the observed mean. A value near one indicates that predictions explain much of the observed variation; a value can be negative if the model performs worse than predicting the sample mean.

The group paper reports dew-point R^2 of approximately 0.66 and MAE of approximately 0.58 °C, and an approximate QNH error of 1.8 hPa. The QNH metric type is not fully specified in the summary, and no separate RH metric is reported.

For evaluation metrics, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3.5 Data analysis

Initial analysis should summarise observation count, date range, sampling cadence, missingness, duplicates, ranges and distributions. Time-series plots reveal gaps and abrupt changes; summary statistics and histograms can reveal suspicious values.

After prediction, residuals should be calculated as actual minus predicted. A residual mean near zero does not rule out systematic errors in particular seasons or pressure ranges. Residuals should be plotted against time, predicted value and relevant weather conditions.

The group paper provides summary metrics rather than full residual data. This thesis therefore does not invent residual distributions, confidence intervals or season-specific scores. It describes the analysis that should be performed when underlying prediction records are available.

For data analysis, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and

prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3.6 Ethics, security, and responsible AI

Meteorological observations are environmental data, but responsible handling still matters. Data source and access conditions should be documented. API credentials, database secrets and passwords must not be embedded in the thesis or a public repository.

Forecasts should be labelled as model-generated estimates and not presented as official operational instructions. Aviation decisions require authorised meteorological products and established procedures. Limitations should be visible when inputs are stale or model performance degrades.

Generative AI and automated tools must be disclosed according to university policy. The student remains responsible for verifying factual claims, equations, citations, code and results, and for ensuring the thesis accurately represents work completed.

For ethics, security and responsible ai, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

3.7 Reproducibility

Reproducibility requires more than a saved model file. Preserve the data-preparation script, feature definitions, target alignment, training configuration, evaluation script and software environment. Record exact training and test periods.

Record software versions and model parameters. If a live source changes schema or reporting conventions, validate and document the change. A model should have a version identifier and should not be silently overwritten by retraining.

The paper describes automated retraining and real-time validation. Each retraining event should create a new version, retain the previous model and compare candidate performance on later unseen observations.

For reproducibility, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component must generate three-hour-ahead QNH and dew-point forecasts, calculate RH from predicted temperature and dew point, and preserve metadata for verification. It must integrate with storage and the wider presentation layer.

Functional requirements specify the outputs and processing actions. Non-functional requirements include reliability, traceability, maintainability, reproducibility and safe failure. A numerically accurate model can still be unusable if its forecast target time or data freshness is unclear.

For requirements, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

4.2 Component architecture

The logical architecture includes data access, validation, preprocessing, feature engineering, model inference, RH calculation, persistence and verification. Data access retrieves observations; validation checks fields, units and timestamps; feature engineering creates a model-ready input; and the model layer predicts QNH and dew point.

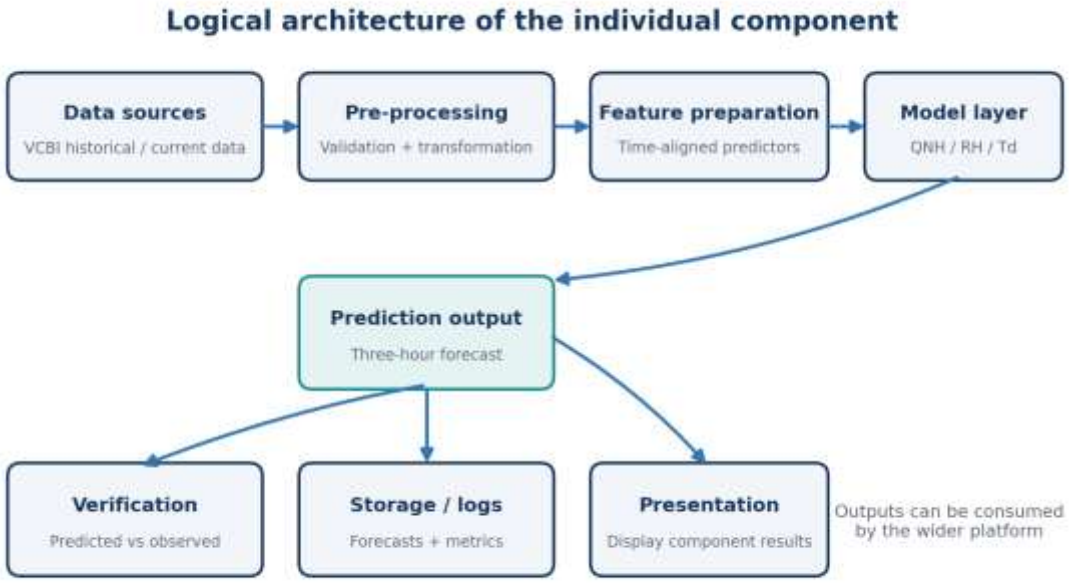


Figure 4-Logical architecture of the individual component

The RH calculation uses predicted dew point and a temperature forecast aligned to the same target time. Persistence stores outputs and metadata. Verification later matches predictions to observations and computes errors.

The group paper describes FastAPI in the wider backend and a dashboard for observations, predictions and verification. Exact API routes and payloads should be taken from the implemented application rather than invented in this report.

For component architecture, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

4.3 Design alternatives

One alternative is to train an independent RH model. The documented design calculates RH from predicted temperature and dew point, supporting physical consistency. The limitation is that errors in either input propagate into RH, and the equation has an applicable domain.

Another option is a multi-output model for QNH and dew point. The paper describes separate models, allowing independent tuning and evaluation. Cross-target dependencies are not automatically learned unless represented through features or downstream calculations.

A random train/test split is less appropriate for forward forecasting than a chronological split because the latter better reflects deployment. The chronological design is retained.

Periodic retraining is operationally manageable, but model promotion should depend on data-quality checks and evaluation criteria. A scheduled training task does not by itself prove that a candidate is better.

For design alternatives, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

4.4 Detailed design

Target generation pairs an input at issue time t with the target observation at t+3 hours. Timestamp-based joins are preferred where observations may be missing. Missing targets should be excluded and counted.

Feature generation includes cyclic time encodings, temperature-dew-point difference, pressure movement and rolling averages. Each feature must be reproducible from past observations. The window definition and treatment of partial windows should be documented.

Inference validates the feature vector against the training schema. Missing features or unexpected units should produce a controlled error rather than an untraceable prediction. Each result should include variable, value, unit, issue time, target time and model version.

Verification matches a forecast to the observation at its target time. If the observation is delayed, the forecast remains pending. Source corrections should be tracked so revised verification is distinguishable from the original record.

For detailed design, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

4.5 Professional practice

Professional practice requires consistent units, clear definitions, documented assumptions and secure data access. The final implementation should confirm that QNH is in hPa and temperature values are in °C throughout the pipeline.

Software engineering practices include modular code, automated tests, source control, configuration management and review of data-pipeline changes. Model versions should be retained and compared using the same evaluation method.

Operational claims must remain proportional to the evidence. The available summary does not establish aviation certification, safety approval or operational acceptance.

For professional practice, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

5. IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Environment and technology stack

The supplied paper identifies LightGBM for the QNH and dew-point subsystem, SQLite for local data storage, and FastAPI as part of the wider backend. The overall system also includes a React interface and other module-specific tools.

The exact operating system, Python version, package versions and deployment configuration are not stated in the supplied files. These details should be added from the actual development environment rather than inferred.

Implementation units should be separated into data loading, preprocessing, feature engineering, target alignment, training, inference, RH calculation, verification and retraining. This separation improves testability and troubleshooting.

For environment and technology stack, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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5.2 Component implementation

The data loader should connect to the configured SQLite database and retrieve required fields deterministically. Database paths should come from configuration. Training data loading should not write to source tables unless that behaviour is explicitly required.

Preprocessing should produce a clean dataset and a quality summary. Record counts loaded, transformed, removed and excluded. This makes it possible to explain differences between training runs.

Feature engineering should be shared between training and inference to avoid training-serving skew. Cyclic features use a consistent period, rolling windows use past observations only, and pressure momentum uses a documented lag.

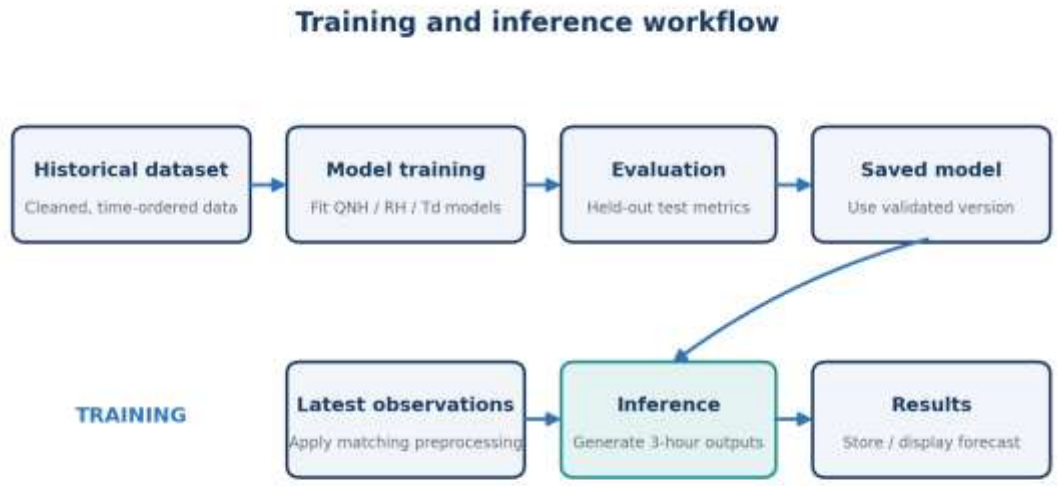


Figure 5-Training and inference workflow

Separate LightGBM regressors are trained for QNH and dew point. Training configuration should include target, forecast horizon, features, training interval and hyperparameters. Save each model with metadata and evaluation output.

The RH function receives predicted dew point and predicted dry-bulb temperature for the same target time. It applies the specified equation and checks for non-finite outputs. If RH is outside plausible bounds, flag the result and investigate the inputs rather than silently clipping it.

The prediction interface should return a structured result including variable, value, unit, issue timestamp, target timestamp, model version, input-data timestamp and RH calculation method when applicable.

For component implementation, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

5.3 Integration

The component depends on data acquisition for recent METAR observations and on the temperature model for the dry-bulb forecast required by RH. It provides forecasts for storage and visualisation alongside other airport-weather outputs.

The wider paper describes storage of observations, predictions and verification logs. An integration contract should define variable identifiers, units, timestamp format, model version and error states. The interface should distinguish pending, verified and unavailable forecasts.

Integration tests should use controlled fixtures as well as live feeds. Fixtures should include a complete series, a missing timestamp, duplicate data, out-of-order records and invalid numeric values.

For integration, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

5.4 Testing

Unit tests should cover timestamp parsing, chronological sorting, target alignment, cyclic features, rolling calculations, RH formula outputs and metric calculations. Include boundary cases such as missing values, empty datasets and insufficient history for a rolling window.

Data-leakage tests are essential. For a selected issue time, feature values should remain unchanged if later observations are appended. If earlier features change, future information may be entering the calculation.

Evaluation tests should confirm that the test interval follows the training interval and that target rows are aligned at the required horizon. Metrics should be calculated from equal-length arrays of aligned observations and predictions.

Integration tests should verify that forecasts are stored with correct target timestamps and matched to later observations. Performance tests should measure data loading, feature generation, inference and persistence; the supplied paper does not provide latency figures.

For testing, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

5.5 Deployment and operation

The operational workflow ingests new observations, validates them, generates features and predicts QNH and dew point. RH is calculated when temperature and dew point forecasts are valid and aligned. Forecasts are stored and made available to the dashboard.

The paper describes automated retraining and live validation. Retraining and model promotion should be separate: a candidate is trained and evaluated in the background, then promoted only if it meets predefined quality and data requirements.

Monitoring should include data freshness, missing fields, prediction failures, unusual output ranges, verification coverage and recent errors. If the feed fails, the service should report unavailable or stale inputs rather than silently presenting an old forecast as current.

For deployment and operation, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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6 RESULTS AND VALIDATION

6.1 Research results

The results in this chapter come from the supplied group research paper. They are not presented as an independent rerun because the underlying database, prediction files and fitted model artefacts were not included with the supplied documents.

The paper reports a dew-point LightGBM model with R^2 approximately 0.66 and MAE approximately 0.58 °C. This suggests that the model captured some variation in the held-out observations, with average absolute error a little over half a degree Celsius in the reported evaluation.

For QNH, the paper reports an approximate error of 1.8 hPa. The summary does not clearly identify the metric type, so it is not relabelled as confirmed MAE. No separate QNH R^2 is provided in the summary.

RH is calculated from predicted temperature and dew point. The paper does not report a separate RH MAE, RMSE or R^2, so RH accuracy cannot be quantified from the supplied summary alone.

For research results, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

6.2 Technical results

The architecture separates QNH and dew-point prediction and calculates RH through a physical equation. This improves traceability because QNH and dew point are model outputs while RH is formula-derived.

The paper describes UTC sorting, missing-value filling, time-related features, temperature-dew-point difference, pressure movement and three- and six-hour averages. Exact implementation details should be confirmed from source code.

The wider system includes MLOps activities such as automated retraining and real-time verification. These capabilities support maintenance, but their presence does not prove that retraining improves future error.

For technical results, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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6.3 Stakeholder validation

The supplied files do not include a structured user study, stakeholder questionnaire or operational acceptance record for this individual component. No user satisfaction percentage or formal approval is claimed.

An appropriate review would ask meteorological or aviation-domain reviewers to examine variable definitions, units, timestamps, forecast displays and limitations. Reviewers should be

able to identify when a prediction was issued, when it applies, and whether it has been verified.

Future stakeholder validation should record participant role, evaluation task, instrument, consent procedure where applicable and analysis method. Claims based on feedback should only be added after the evaluation is completed.

For stakeholder validation, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

6.4 Integration results

The group paper places this subsystem in a wider system that acquires METAR data, stores observations and predictions, provides backend services and visualises weather outputs. It establishes the intended integration context but does not include component-specific API test logs.

An integration test should show that QNH and dew-point predictions are stored with consistent timestamps and that RH uses temperature and dew point for the same target time. Another test should show that verification matches the exact target timestamp.

Until tests are supported by logs or screenshots, they are acceptance criteria rather than completed tests. This distinction prevents the architecture described in the paper from being mistaken for execution evidence.

For integration results, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

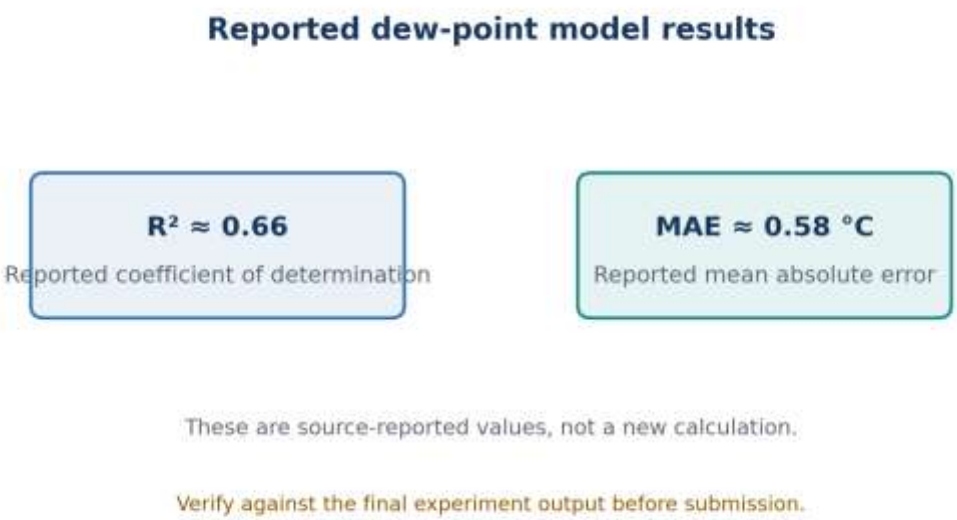
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6.5 Findings summary

The available evidence indicates a reported dew-point R^2 of about 0.66 and MAE of about 0.58 °C. The QNH model has an approximate reported error of 1.8 hPa, but the metric type is not fully specified.

Figure 6 - Reported dew-point model results



The main result is a documented forecasting approach with preliminary numerical performance for two modelled variables and a physically motivated method for RH. Stronger conclusions require a precise QNH metric definition, RH verification, sample counts and season-wise analysis.

For findings summary, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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7 DISCUSSIONS

7.1 Interpretation

The dew-point R^2 of approximately 0.66 indicates that the model explains a meaningful portion of variation in the reported test observations. MAE of approximately 0.58 °C is a more direct description of average absolute error, but it does not mean every prediction lies within that distance.

The QNH error of approximately 1.8 hPa should be interpreted cautiously because the summary does not fully specify the metric. The final result table should state metric, unit, sample count and evaluation interval.

Derived RH offers consistency between predicted moisture and temperature, but it does not automatically guarantee accuracy. Errors in temperature and dew point propagate nonlinearly through the formula, so RH needs its own evaluation.

A model's performance at a three-hour horizon cannot be generalised automatically to six-hour forecasts, other airports or every monsoon phase. Conclusions should remain limited to the reported evaluation.

For interpretation, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

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7.2 Comparison with related work

The group paper positions the component alongside machine-learning approaches in weather forecasting and aviation. Its airport-specific focus is consistent with the need for local information, but the supplied paper does not provide a controlled comparison against a numerical weather prediction baseline for the same observations.

Results from other group modules are not directly comparable. Wind errors are expressed in knots, cloud and visibility may use classification accuracy, and QNH/dew point use regression metrics. Targets and units differ.

A future baseline comparison should include persistence, predicting the target from the latest observed value, and possibly a seasonal baseline. All models should use identical issue timestamps, target timestamps and chronological test periods.

For comparison with related work, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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7.3 Research contribution

The contribution is a defined airport-level component that predicts QNH and dew point three hours ahead and derives RH from predicted atmospheric variables. The separation between learned outputs and formula-derived RH makes the dependency explicit.

The temporal processing and verification design is also important. UTC ordering, horizon-specific target alignment, past-only features and future-observation matching are essential for valid forecast evaluation.

The contribution is not a new atmospheric equation or a universally novel algorithm. It is the application, integration and evaluation of established techniques in the VCBI monsoon-focused system.

For research contribution, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

7.4 Product contribution

The component can supply a compact set of forecast values to an airport-weather interface. Outputs are more useful when accompanied by issue time, target time, unit, model version and verification status.

Verification can support maintenance by identifying error trends and data problems. A sudden rise in error might reflect a weather regime shift, an input-data change, a timestamp bug or model degradation; monitoring should examine both model and pipeline health.

For product contribution, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

7.5 Limitations and validity threats

The supplied files contain a summary paper rather than the full experimental artefacts. The row-level dataset, fitted models, exact LightGBM parameters, complete metrics and sample counts are absent, limiting independent reproduction.

The QNH metric is not fully specified and no RH performance score is reported. A complete evaluation requires the precise metric names, sample counts and evaluation interval from the actual experiment logs.

A single chronological 80/20 split can be sensitive to the seasons represented in the test period. Rolling-origin evaluation and year-wise testing would provide stronger evidence of robustness.

Missing-data handling is described only generally. Forward filling, interpolation and row deletion have different effects, particularly during rapid weather changes. The chosen approach should be stated and tested.

RH depends on the temperature forecast as well as dew point. Even a reasonable dew-point forecast can produce poor RH if temperature is biased or misaligned. Both inputs and the derived output require verification.

Point forecasts do not communicate uncertainty. Future work could investigate prediction intervals or quantile regression, with calibration evaluated on held-out data.

The thesis does not establish regulatory approval, operational safety certification or formal acceptance. The component should be treated as a research system unless validation and authorisation are documented.

For limitations and validity threats, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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7.6 Implications

For researchers, the component demonstrates the importance of aligning targets with the forecast horizon and evaluating chronologically. For developers, it highlights the need to separate data preparation, prediction, physical calculation and verification.

For meteorological users, forecasts should be accompanied by units, timestamps and limitations. For maintainers, continuous verification can reveal performance changes, but retraining should follow evidence-based criteria.

For academic reporting, distinguishing source-derived results from future work improves transparency and helps evaluators understand exactly what was demonstrated.

For implications, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

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8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition

The potential value is to provide structured short-term estimates of pressure and moisture-related variables for an airport-focused dashboard. Potential users include research teams and meteorological analysts who need to compare observations, forecasts and verification records. These are potential users, not evidence of completed market validation.

Value depends on accuracy, freshness, interpretability and responsible presentation. A forecast without target time or error history can mislead even if its average test score is acceptable.

The component should be positioned as an analytical aid, not a replacement for official aviation meteorological products. Operational use would require domain review and suitable governance.

For value proposition, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

8.2 Competitive position

Existing weather services and aviation meteorological systems provide broad forecasting capabilities. This project’s focus is a local research workflow joining airport observations, short-term model output, derived humidity and subsequent verification.

A fair competitive assessment would compare persistence forecasts, public weather APIs, official products and other local approaches using the same targets and test interval. It should include operating costs, data availability and maintenance requirements.

The supplied files do not provide a market survey, customer interviews, pricing analysis or cost model. Commercial claims should be supported by additional research and review of data rights.

For competitive position, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

8.3 Sustainability model

A possible sustainability model is an institutionally maintained research service with periodic evaluation, documented data access and a named owner for monitoring. Costs may include data access, compute, storage, software maintenance, evaluation and human review.

Model maintenance is a continuing cost because performance can degrade when data or weather regimes change. Responsibility should be allocated for data quality, retraining approval, monitoring, incident handling and rollback.

Open documentation and reproducible experiments can reduce handover costs. Credentials and restricted data should not be published, while non-sensitive code and evaluation methods can be shared where licensing permits.

For sustainability model, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

8.4 Scalability and compliance

The logical design can be adapted to other airports, but transfer should not be assumed to work without retraining and validation. Airports differ in elevation, climatology, reporting practice and observation availability.

Ownership and reuse conditions should be confirmed with the university, project group and data providers. Third-party libraries and data sources have their own licences and terms.

Compliance considerations include source terms, secure credentials, accurate representation of forecast status and aviation-domain oversight. The component remains a research prototype unless its validation and approval status are established.

For scalability and compliance, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

8.5 Sustainable development

The project may indirectly support resilient infrastructure by exploring data-driven methods for local weather information. This thesis does not measure a direct reduction in incidents, emissions or operational delays.

The sustainability argument should remain modest: the project contributes an academic prototype and repeatable evaluation concept. Claims of measurable environmental or safety benefit would require a separate outcome evaluation.

For sustainable development, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis documents an individual component of the hyper-local aviation weather forecasting and verification system for VCBI. It focuses on three-hour-ahead QNH and dew-point prediction and derives RH from predicted dew point and predicted dry-bulb temperature using the August–Roche–Magnus equation.

The supplied group paper identifies LightGBM for separate QNH and dew-point models, with data stored in SQLite, chronological UTC ordering and engineered time and meteorological features. It reports dew-point R^2 of approximately 0.66 and MAE of approximately 0.58 °C. The QNH model has an approximate reported error of 1.8 hPa, but the metric type is not fully specified.

The central conclusion is that combining data-driven forecasts with a physical humidity relationship is a reasonable design, provided temperature and dew-point forecasts are aligned and verified. The reported metrics are preliminary evidence; more reproducible experiments are needed before claims about all monsoon conditions or operational suitability.

For conclusion, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

9.2 Objective achievement

The objective of documenting time-ordered data preparation is addressed through UTC sorting, missing-value handling and target alignment. The exact implementation details must be confirmed from the source code.

The objective of documenting temporal and meteorological features is addressed by the reported cyclic time encodings, temperature-dew-point difference, pressure movement and rolling averages. Further details should be included only when supported by implementation records.

The QNH and dew-point modelling approach is documented using LightGBM and a three-hour target horizon. The available metrics provide initial evidence, although full test dates and sample counts are not in the supplied files.

The RH objective is addressed through the August–Roche–Magnus equation. Full RH evaluation, a precise QNH metric definition, season-wise testing and independent reproduction remain future work.

For objective achievement, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

9.3 Individual contribution

The individual contribution is the focused analysis and documentation of the QNH, dew-point and RH forecasting pathway, including its horizon, feature categories, model choice, derived-variable calculation, evaluation interpretation and limitations.

Shared infrastructure, other prediction modules and the wider dashboard are acknowledged as group context. The final submission should identify which source files, model experiments, APIs and tests were personally developed.

For individual contribution, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

9.4 Future directions

Reproduce the evaluation from the actual dataset and saved predictions. State the test interval, number of samples, LightGBM configuration and full MAE, RMSE and R^2 for each modelled target.

Evaluate RH against observed RH at aligned target timestamps, reporting MAE, RMSE, bias and sample count. Examine sensitivity to temperature and dew-point errors.

Use rolling-origin and season-wise evaluation to assess robustness across monsoon periods. Compare with persistence and seasonal baselines using identical test intervals.

Document missing-value handling and run sensitivity analysis for gaps, duplicates, late observations and schema changes.

Explore uncertainty estimates and monitoring rules. Any prediction intervals or drift alerts should be evaluated on held-out data.

Proceed with deployment only after domain review and appropriate authorisation. Model outputs should remain labelled as estimates and complement, not replace, official aviation weather products.

For future directions, the central methodological requirement is traceability. Each input, transformation and output should be associated with a documented definition so that another researcher can determine how the forecast was produced. This is especially important when results are later compared across different seasons or model versions.

The evidence available in the supplied group paper supports describing the intended approach, but it does not support inventing unreported test statistics or implementation details. Where the full dataset or source code is needed, the final submission should add the verified information directly from those artefacts.

A practical evaluation should retain a record of the observation count, target period, unit conventions and any exclusions. These details help explain the scope of the result and prevent a single summary metric from being interpreted as proof of performance under every possible weather condition.

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GLOSSARY

Term	Definition
Actual observation	A measured or reported meteorological value associated with a specific observation time.
Chronological split	A train/test division in which the test interval occurs after the training interval.
Concept drift	A change over time in the relationship between input features and the target.
Dew-point depression	Difference between dry-bulb temperature and dew-point temperature.
Forecast issue time	Time at which a forecast is generated.
Forecast target time	Future time to which the forecast applies.
Feature engineering	Transformation of raw observations into model input variables.
LightGBM	Gradient-boosting framework that uses decision trees for supervised learning.
MAE	Mean of absolute differences between predictions and actual values.
METAR	Coded aerodrome meteorological observation report.
Model version	Identifiable trained model artefact and associated configuration.
QNH	Aviation altimeter setting referenced to mean sea-level pressure.
Relative humidity	Percentage expressing actual vapour pressure relative to saturation vapour pressure at the same temperature.
RMSE	Square root of mean squared difference between predictions and observations.
R^2	Coefficient of determination comparing model predictions with variation in the target.
TAF	Terminal aerodrome forecast.
Temporal leakage	Use of information that would not be available at forecast issue time.
UTC	Coordinated Universal Time used as the canonical timeline.

APPENDIX A: DATA DICTIONARY AND DATA QUALITY SPECIFICATION



Figure 7-Data quality process

This is a working data dictionary based on the supplied paper. Confirm the actual SQLite schema before treating every optional field as present in the implemented system.

Table 1 - Dataset fields and their uses

Field	Meaning / use	Unit / format	Status
timestamp_utc	Observation time and temporal alignment	UTC timestamp	Required concept
qnh	Observed target and lag feature	hPa	Reported variable
temperature	Dry-bulb temperature used in RH calculation	°C	Required RH input
dewpoint	Observed target and lag feature	°C	Reported variable
relative_humidity	Observed RH for validation if available	%	Confirm availability
source	Origin of observation	Text	Recommended
quality_flag	Marks suspect or missing observations	Categorical	Recommended
issue_time_utc	Forecast creation time	UTC timestamp	Recommended

target_time_utc	Time forecast describes	UTC timestamp	Required for verification
prediction_value	Forecast value	Variable-specific	Required
actual_value	Matched observation	Variable-specific	Required for verification
absolute_error	Absolute difference between forecast and actual	Variable-specific	Calculated

Ingestion checks

Confirm the database connection and required tables. The process should fail clearly if the path is invalid or a required column is renamed. It should not silently create an empty replacement database and then appear to operate normally.

Parse timestamps and sort records. Report duplicate counts and missing intervals. If hourly cadence is expected, inspect differences between adjacent timestamps. A gap should not automatically be treated as an ordinary adjacent row.

Check numeric types and plausible ranges using documented criteria. Extreme observations may represent real weather, so flags should trigger review rather than automatic deletion unless the rule is justified.

Record row counts before and after each cleaning step so that dataset changes between training runs can be explained.

Missing-value policy

The group paper states that missing data are filled but does not specify the exact method. Document the policy for each variable. A short gap may be handled differently from a long gap, and forward filling can create stale values during rapid changes.

Target values should not be invented to create labels. If the t+3 target observation is missing, exclude the example or apply a justified strategy. Filling a target with an earlier value can artificially reduce apparent error.

Summarise missingness by variable and period. If gaps cluster in a season or outage, the retained sample may not represent all operating conditions.

Provenance and versioning

Each training run should identify a reproducible database snapshot or query interval. Where retention rules allow, preserve the historical sample used for the model.

Record code commit, package versions, model parameters, training period and test period. Retraining should create a new model version and preserve the previous model for comparison or rollback.

If a source observation is corrected, record which version was used for verification or retain the correction timestamp. This helps explain metric changes.

APPENDIX B: FEATURE ENGINEERING SPECIFICATION

Table 2-Engineered feature groups

Feature group	Example definition	Purpose	Leakage control
Cyclic hour	$\sin(2\pi h/24)$, $\cos(2\pi h/24)$	Represent daily cycle	Issue timestamp only
Cyclic seasonal phase	Sine/cosine of annual phase	Represent annual patterns	Calendar time only
Lagged QNH	QNH at prior time steps	Pressure persistence	Past values only
Pressure momentum	Difference between current and prior QNH	Rising/falling trend	Uses prior observations
Temperature-dewpoint spread	$T - Td$	Moisture proximity to saturation	Issue-time observations only
Three-hour rolling mean	Mean over defined prior window	Short-term variation	No future rows
Six-hour rolling mean	Mean over longer prior window	Broader local trend	No future rows
QNH target +3 h	Observed QNH at t+3 h	Supervised target	Never an input
Dew-point target +3 h	Observed dew point at t+3 h	Supervised target	Never an input

Cyclic encoding

Hour-of-day is cyclic: the end of one day is adjacent to the beginning of the next. A sine-cosine pair represents the cycle without a discontinuity at midnight. The period and treatment of fractional hours should be consistent between training and inference.

Seasonal features may use month or a continuous annual phase. Month-based encoding is simple but treats days in a month similarly. The chosen representation should be documented and reused consistently.

Lag and rolling features

Lag features expose recent measurements to the model and can capture persistence. Each lag should be defined relative to elapsed time when timestamps may be missing. A feature called “three hours ago” should not mean merely three rows earlier if gaps exist.

Rolling averages summarise recent conditions. Define the window duration, minimum observation count and treatment of incomplete windows. Partial windows can create missingness at the beginning of the series and should be handled consistently.

Pressure momentum can be a difference between current and earlier pressure. The direction and unit must be documented; a difference between QNH values remains in hPa unless a normalisation is explicitly applied.

Leakage prevention

Leakage can occur when rolling means are centred, future values fill earlier gaps or targets are misaligned. Each can produce deceptively favourable scores.

A useful automated test computes features for an issue time, appends later observations and recomputes the earlier features. The earlier feature vector must remain unchanged. Another test verifies that each target timestamp is exactly three hours after the issue timestamp.

Feature selection and tuning must not use the final test interval. If test results are repeatedly used to choose model settings, that interval becomes part of development and a new later evaluation period is needed.

APPENDIX C: EVALUATION FORMULAS AND INTERPRETATION

Mean Absolute Error

$$\mathbf{MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|}$$

Here y_i is observed, $\hat{y}_i$ is predicted and n is the number of aligned pairs. MAE has the same unit as the target.

Root Mean Squared Error

$$\mathbf{RMSE = \sqrt{[(1/n)\ \Sigma\ (y_i - \hat{y}_i)^2]}}$$

RMSE penalises large errors more heavily than MAE and should be interpreted alongside it.

Coefficient of determination

$$\mathbf{R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2\ /\ \Sigma(y_i - \bar{y})^2]}$$

$\bar{y}$ is the mean observed value. R^2 is not the percentage of forecasts that are correct and can be negative.

RH equation

$$\mathbf{RH = 100 \times \exp[(17.67Td/(Td + 243.5)) - (17.67T/(T + 243.5))]}$$

Td and T are in °C. The implementation should validate units, input alignment and output finiteness.

Table 3-Evaluation metrics for forecast verification

Target	N	MAE	RMSE	R^2	Bias	Evaluation interval
QNH	[from logs]	[fill]	[fill]	[fill]	[fill]	[UTC interval]
Dew point	[from logs]	0.58 °C reported	[not supplied]	0.66 reported	[fill]	[UTC interval]
Derived RH	[from logs]	[not supplied]	[not supplied]	[not supplied]	[fill]	[UTC interval]

Do not submit placeholders as completed empirical results. Replace them with values generated from actual prediction and observation files.

APPENDIX D: SOFTWARE TEST CASE CATALOGUE

Table 4-Component test cases

ID	Test	Setup	Expected result
D-01	Valid database connection	Required table exists	Rows load
D-02	Database unavailable	Invalid path	Controlled error; no false forecast
D-03	Required column missing	Remove qnh from fixture	Schema validation fails
D-04	Timestamp parsing	Valid and invalid timestamp strings	Valid rows parsed; invalid flagged
D-05	Out-of-order records	Shuffle fixture rows	Sorted chronologically
D-06	Duplicate timestamps	Duplicate key present	Policy applied and counted
D-07	Missing target time	Remove t+3 observation	Pair excluded and counted
D-08	Rolling feature warm-up	Series shorter than window	Missing state handled
D-09	Midnight cyclic features	23:00 and 00:00	Cyclic encoding consistent
D-10	Normal RH calculation	Finite T and Td	Finite result returned
D-11	Missing temperature	Temperature unavailable	RH marked unavailable
D-12	Td exceeds T	Inconsistent input	Warning or validation error
D-13	Prediction metadata	Generate forecast	Timestamps, unit and version stored
D-14	Verification match	Forecast and actual same target time	Error calculated
D-15	Timestamp mismatch	Actual from another hour	Incorrect match rejected
D-16	Metric length mismatch	Arrays differ in length	Evaluation fails clearly
D-17	Retraining versioning	Train candidate	New version stored; old retained
D-18	Candidate fails quality gate	Poor validation performance	Candidate not promoted
D-19	Stale observation	Latest input too old	Status indicates stale data
D-20	API unavailable	Model service unreachable	Clear error; no fabricated value

For every executed test, record test identifier, date/time, code version, environment, input fixture, expected and actual results, pass/fail status and evidence location. These are recommended acceptance tests, not claimed as passed unless execution logs are available.

APPENDIX E: MODEL OPERATIONS AND RETRAINING PROCEDURE

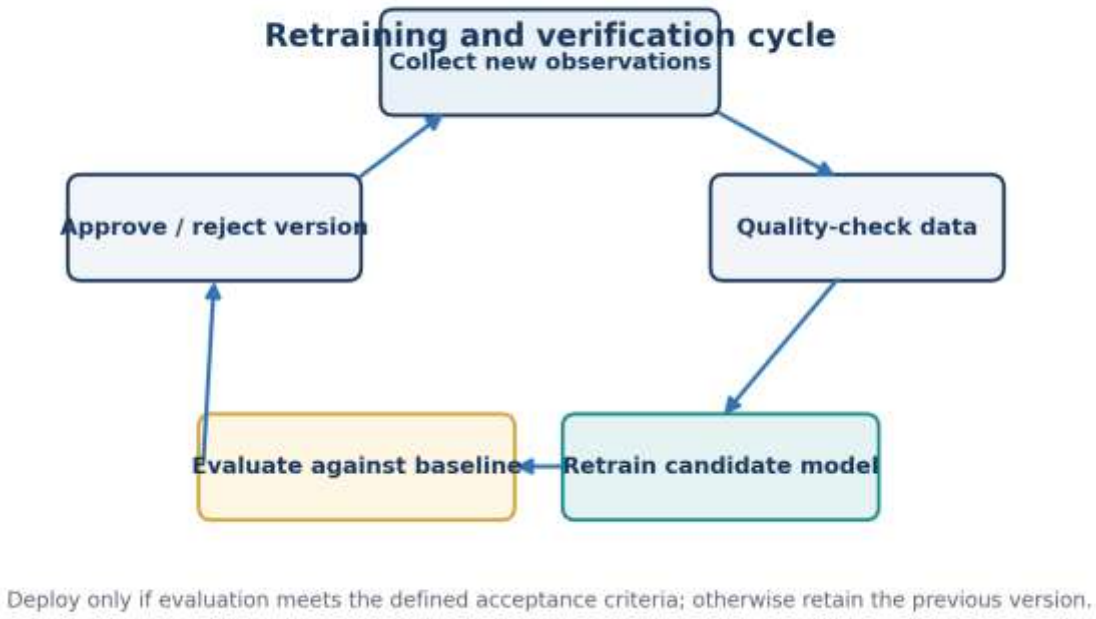


Figure 8-Retraining and verification cycle

Scheduled data refresh

The refresh task retrieves observations and validates the expected schema. It records retrieval time, source status and rows added or updated. If the source is unavailable, it should leave the last valid data intact and alert the operator.

After ingestion, a quality report checks freshness, missing fields, timestamp order, duplicates and value anomalies. Inference should be blocked or flagged when required inputs are absent or stale. The freshness threshold should be configured and documented.

Candidate model training

A scheduled retraining process prepares eligible historical data and creates a candidate model. It records training interval, features, target alignment, validation interval and configuration. Candidate evaluation must use data later than the training period.

The active model remains available while the candidate is evaluated. Promotion requires passing data-quality checks and predefined performance criteria. A completed training job alone is not evidence of improved quality.

If the candidate fails, retain the active version and record why. If it passes, register a new version and preserve a rollback path. Model changes should be visible in verification history.

Live verification and drift monitoring

Each forecast is stored with issue and target times. When the target observation arrives, the verification process joins it by timestamp and station where applicable.

Monitoring summaries should include MAE, RMSE, bias, number of verified forecasts and verification coverage. A favourable metric based on very few forecasts should not be treated as equally reliable as one based on a larger sample.

Potential drift signals include sustained error increases, changed input distributions, rising missingness or a change in observation cadence. Investigate pipeline defects before retraining on potentially corrupted data.

Incident and rollback

When a model produces implausible values or errors increase, mark affected output and alert the maintainer. Check source freshness, feature calculations, model version, target alignment and recent weather conditions.

Rollback restores the last accepted version and records the reason. Previously issued forecasts should not be silently rewritten. If a source observation is corrected, preserve the audit trail for the revised verification.

APPENDIX F: RISK REGISTER AND MITIGATION

Table 5-Risk register for the individual component

Risk	Potential impact	Mitigation	Residual concern
Timestamp misalignment	Wrong observation compared	UTC, exact target join, tests	Source timestamps may be wrong
Temporal leakage	Overstated test score	Past-only features, chronology, tests	Pipeline changes need retesting
Missing or stale data	Unreliable inputs	Freshness checks and controlled failure	Short gaps still affect accuracy
Monsoon regime shift	Error increases	Verification, seasonal analysis, retraining gates	Rare events remain difficult
Temperature forecast error	RH error increases	Evaluate temperature and RH inputs	Nonlinear error interaction
Unclear QNH metric	Misleading interpretation	Recompute full metrics from logs	Needs original prediction records
No RH metric	Unknown RH accuracy	Compare derived RH with observed RH	Reference quality must be checked
Model overwrite	No rollback path	Versioned model artefacts	Retention costs
API/database failure	Forecast unavailable or stale	Health checks and status display	Infrastructure dependency
Overstated safety claims	Misuse of output	Clear labels and domain review	Requires governance

Risk governance

Review risks whenever the source, feature pipeline, model, forecast horizon or intended user changes. Each high-priority risk should have an owner and review date. Risk management does not eliminate machine-learning failure modes; it reduces their likelihood and makes failures easier to detect and recover from.

The system should define how incidents are communicated and how it returns to a known valid state. The risk register should be updated using real incident and verification evidence rather than treated as a one-time academic table.

APPENDIX G: REPRODUCIBILITY CHECKLIST

☐ 1. Confirm the actual SQLite database filename and schema.

☐ 2. Record data source, station identifier, date range and sampling cadence.

☐ 3. Document UTC parsing and duplicate policy.

☐ 4. Document missing-value handling separately for features and targets.

☐ 5. Verify target timestamps are exactly three hours after issue timestamps.

☐ 6. List every feature with definition, unit and lookback window.

☐ 7. Confirm no future observation contributes to input features.

☐ 8. Record LightGBM version and model parameters.

☐ 9. Record training, validation and test intervals.

☐ 10. Preserve model artefacts and prediction files.

☐ 11. Calculate MAE, RMSE and R^2 for QNH and dew point.

☐ 12. Clarify the metric represented by the approximate 1.8 hPa QNH error.

☐ 13. Evaluate derived RH against observed RH if available.

☐ 14. Record sample counts and excluded target rows.

☐ 15. Include residual plots and season-wise metrics where feasible.

☐ 16. Run unit, integration and leakage tests.

☐ 17. Record software versions and source-control commit.

☐ 18. Document retraining and model promotion criteria.

☐ 19. Disclose AI assistance according to university policy.

☐ 20. Update Word table of contents and page fields.

☐ 21. Remove placeholders before submission.

APPENDIX H: PSEUDOCODE FOR THE FORECASTING WORKFLOW

```
load observations from SQLite
validate required columns and timestamp format
convert timestamps to UTC and sort chronologically
inspect duplicates and missing values
apply documented data-quality policy
for each issue timestamp t:
    calculate features using observations available at or before t
    align QNH target to observation at t + 3 hours
    align dew-point target to observation at t + 3 hours
    retain only valid target pairs
split samples chronologically
fit LightGBM QNH regressor on training interval
fit LightGBM dew-point regressor on training interval
predict QNH and dew point on the test interval
calculate MAE, RMSE and R-squared for each target
save predictions, actuals, timestamps and model metadata
for each live issue time:
    validate latest observations and feature schema
    predict QNH and dew point at issue time + 3 hours
    obtain temperature forecast for the same target time
    if temperature and dew point are valid:
        calculate RH using the August-Roche-Magnus equation
    else:
        mark RH unavailable
    store forecast values and model version
when target observation becomes available:
    match observation by target timestamp and station
    calculate variable-specific errors
    update monitoring summaries
    investigate drift or data-quality alerts
```

This pseudocode is a design aid. Exact table names, API routes and scheduler configuration should be taken from the project source code before it is represented as the implementation.

The forecast record should remain immutable after verification. Later observations calculate errors but should not overwrite the original prediction. This separation supports auditability and fair comparison across model versions.

APPENDIX I: VERIFICATION LOG TEMPLATE

Table 6-Forecast records and verification fields

Issue time UTC	Target time UTC	Variable	Predicted	Actual	Error	Absolute error	Model version
[timestamp]	[timestamp]	QNH	[hPa]	[hPa]	[actual-predicted]	[value]	[version]
[timestamp]	[timestamp]	Dew point	[°C]	[°C]	[actual-predicted]	[value]	[version]
[timestamp]	[timestamp]	RH	[%]	[%]	[actual-predicted]	[value]	[derived]

Verification interpretation

Signed error indicates whether a forecast is high or low, while absolute error measures the magnitude without regard to direction. Both should be monitored. Results should be grouped by variable and reported with the number of valid pairs.

If only some forecasts can be verified because observations are missing, report verification coverage. Otherwise, a metric may appear favourable simply because difficult periods are underrepresented.

APPENDIX J: EXTENDED DISCUSSION OF QNH FORECASTING

Meaning and relevance of QNH

QNH is an aviation altimeter setting referenced to mean sea-level pressure. It supports interpretation of altitude when used according to the applicable aviation procedures. This thesis does not redefine those procedures.

Forecasting QNH requires a consistent target definition. The model should predict the QNH value reported or derived at the target time using the same convention as historical data. If observations are rounded or have different precision, the evaluation should document that limitation.

Pressure often exhibits persistence over short intervals but can change with weather systems. Lagged values and pressure momentum may be useful predictors; their value should be tested empirically rather than assumed.

QNH-specific data concerns

Pressure values can be affected by unit conversion, parsing errors and reporting conventions. Use one standard unit and validate conversions with unit tests.

Observation freshness matters because pressure may change between reports. Record the age of the latest valid observation and flag forecasts based on stale inputs.

The approximate 1.8 hPa error should not be compared with a tolerance or treated as a confirmed MAE unless the original evaluation output establishes the metric and the relevant operational standard.

QNH evaluation recommendations

Report MAE and RMSE in hPa, R^2 where meaningful, mean signed error, sample count and exact UTC test interval. Include a persistence baseline and state whether the model improves on it.

Inspect errors during rapid pressure changes separately from stable periods. Define the rapid-change criterion and distinguish planned analysis from exploratory analysis.

Compare performance by season and pressure range where sample sizes permit. Such analysis can reveal whether aggregate performance hides weak periods.

APPENDIX K: EXTENDED DISCUSSION OF DEW-POINT FORECASTING

Meaning and relevance

Dew point is the temperature at which air would become saturated under specified pressure and moisture-content assumptions. It is useful because it describes atmospheric moisture in temperature units and can be compared with dry-bulb temperature.

A small temperature-dew-point gap often accompanies high RH, but dew point alone does not determine visibility or fog. Cooling, mixing, wind, terrain and droplet formation also matter.

At an airport, dew-point prediction is useful as an input to broader awareness, but it is not a complete visibility model or an automatic operational decision rule.

Interpreting the reported result

The paper reports R^2 about 0.66 and MAE about 0.58 °C. MAE means the average absolute difference was about 0.58 °C over the reported sample; it does not mean every prediction fell within that distance.

R^2 around 0.66 indicates improved squared-error performance relative to an observed-mean baseline under the standard definition. It is not the percentage of forecasts that are correct.

The result would be more informative with RMSE, bias, absolute-error distribution, sample count, test dates and seasonal metrics. Scatter plots and residual time series can reveal systematic errors.

Feature and target design

Temperature-dew-point spread is physically meaningful but must be computed from values available at issue time. Using future target values to calculate input features would invalidate the evaluation.

Seasonal features represent recurring annual patterns, while lags and rolling means represent recent persistence. Lookback length is empirical: short windows may miss broader patterns and long windows may smooth rapid transitions.

Target alignment should use timestamps. A three-row shift is not necessarily three hours when observations are irregular. Document any tolerance for imperfect sampling.

APPENDIX L: EXTENDED DISCUSSION OF RELATIVE HUMIDITY

Physical relationship

RH depends on temperature because saturation vapour pressure changes with temperature. If water-vapour content stays approximately constant while air warms, RH generally decreases; if air cools toward dew point, RH generally increases.

The group paper uses the August–Roche–Magnus equation to calculate RH from predicted dew point and dry-bulb temperature. Physical consistency does not guarantee forecast accuracy if either input is inaccurate.

The constants in empirical vapour-pressure approximations depend on the formulation and applicable temperature range. This report reproduces the constants stated in the group paper; reconcile code and text if the implementation differs.

Error propagation

RH error can arise from dew-point error, temperature error, rounding, timestamp mismatch or implementation error. Because the formula is nonlinear, the effect of a one-degree error varies with conditions.

A diagnostic experiment can compare RH derived from two predicted inputs with calculations using one observed input at a time. Those latter cases help isolate error sources but are not deployable forecasts because they use information unavailable at issue time.

The operational result is the case in which both temperature and dew point are forecast values for the same target time. Diagnostic results should be labelled clearly.

Validation protocol

Obtain observed RH or calculate a reference RH from trusted observed temperature and dew point with the same equation. If the source provides RH directly, document its precision and source convention.

Align predicted RH with reference RH at the exact target timestamp. Calculate MAE, RMSE, bias and sample count. Investigate values outside plausible bounds and stratify by temperature-dew-point spread and season where possible.

Do not claim RH accuracy until the protocol is completed on a clearly defined evaluation interval.

APPENDIX M: LIMITATIONS AND THREATS TO VALIDITY

Construct validity

The targets must be defined consistently with observations and intended use. A QNH field that changes unit or meaning between sources can invalidate evaluation. RH depends on both the formula and its input definitions.

Internal validity

Temporal leakage, wrong target alignment, duplicates or preprocessing before the split can bias results. Use timestamp-based joins and automated leakage tests.

External validity

Results apply to the dataset and airport context described in the group paper. They do not automatically generalise to other airports, seasons or forecast horizons.

Statistical conclusion validity

A metric based on an unknown or small sample may be unstable. Report sample counts, uncertainty where feasible and consistent periods for all model comparisons.

Operational validity

Average error can hide failures during rare or high-impact weather. Operational readiness requires domain review, data freshness controls and clear boundaries.

Reproducibility validity

The supplied files do not include full data, code or model artefacts. Independent replication requires a reproducibility package where data rights permit.

Mitigation requires pre-specified evaluation design, past-only features, versioned datasets and models, transparent reporting and multiple forms of evidence. If test results are used to modify the model, a later independent evaluation period is needed.

APPENDIX N: FUTURE EXPERIMENT PLAN

Table 7-Planned model experiments

Experiment	Purpose	Method	Outputs
Persistence baseline	Measure improvement beyond last value	Predict target from latest observation	MAE, RMSE, R^2, bias
Rolling-origin evaluation	Assess robustness over time	Train on past, test later, roll forward	Fold and aggregate metrics
Monsoon-stratified analysis	Assess seasonal variation	Group test predictions by documented season	Metrics and sample counts
Missing-value sensitivity	Assess data-cleaning impact	Compare policies on same test periods	Metric changes and retained rows
Feature ablation	Measure feature-group contribution	Remove one group at a time	Metric change and runtime
RH validation	Quantify derived humidity error	Compare derived RH with aligned reference	MAE, RMSE, bias, sample count
Drift monitoring	Develop evidence-based alerts	Define threshold on validation data	Alert quality and detection delay
Retraining gate	Avoid harmful model promotion	Compare candidate and active model	Decision and rationale

Experiment governance

Before each experiment, record research question, data interval, inclusion criteria, features, model parameters and metrics. This reduces selective reporting and supports reproduction.

Use the final test interval only after model choices are fixed. If the same interval is repeatedly used for tuning, it becomes a development set and a new later evaluation interval is needed.

APPENDIX O: RESPONSIBLE PRESENTATION OF FORECASTS

Recommended display fields

Display variable name, value, unit, issue timestamp, target timestamp, input-data freshness, model version and verification status. Identify RH as derived from predicted temperature and dew point.

Colour-coded warnings may be useful, but thresholds must come from documented domain review. A warning should explain its trigger and should not imply a certified operational instruction.

Distinguish “forecast generated,” “awaiting verification,” “verified” and “forecast unavailable.” If the latest observation is stale, display its age.

Communication and user training

Users should understand that forecasts are estimates with measurable error and that past performance does not guarantee future accuracy. Explain the three-hour horizon, units and verification status.

Training should cover stale data, unverified forecasts, error metrics and when to consult official meteorological information. Model output should not be presented as a substitute for professional judgement.

APPENDIX P: FINAL SUBMISSION QUALITY CHECK

☐ 1. Replace group ID placeholder with the confirmed identifier.

☐ 2. Confirm degree title and specialisation wording required by SLIIT.

☐ 3. Confirm student name and ID against official records.

☐ 4. Add supervisor names as required.

☐ 5. Update the abstract after final experiments and keep within the required word limit.

☐ 6. Check implementation claims against source code.

☐ 7. Clarify the metric represented by the approximate 1.8 hPa QNH error.

☐ 8. Add RH evaluation if aligned observations are available.

☐ 9. Add exact test dates, sample counts and full metrics.

☐ 10. Insert genuine architecture diagrams and result plots where available.

☐ 11. Verify every in-text citation against the reference list.

☐ 12. Check bibliographic details against original publications.

☐ 13. Update table of contents, figure list, table list and page fields.

☐ 14. Remove template guidance and unresolved placeholders.

☐ 15. Check margins, font, spacing and heading hierarchy.

☐ 16. Proofread spelling, equations and units.

☐ 17. Confirm AI-use disclosure meets university policy.

☐ 18. Remove unsupported safety or certification claims.

☐ 19. Back up scripts, data, model artefacts and evaluation outputs.

☐ 20. Obtain supervisor feedback and follow final submission instructions.

APPENDIX Q: DATA FLOW AND FORECAST RECORD DESIGN

Observation record

An observation record should represent a value reported at a specific station and timestamp. A minimum record includes timestamp, variable values and source provenance. When multiple stations or sources are used, the station identifier and source should be preserved.

The ingestion process should avoid silently replacing one source observation with another. If a correction or duplicate arrives, the database should record how it was resolved. Data-quality flags allow downstream processes to distinguish an accepted observation from one requiring review.

Forecast record

A forecast record represents a prediction made at an issue time for a target time. It should include target variable, predicted value, unit, issue timestamp, target timestamp, model version and input-data freshness. For RH, it should also identify the temperature and dew-point forecasts used in the calculation.

Forecast records should be immutable where practical. Verification adds actual values and error fields or creates a related verification record. It should not replace the prediction value with an actual value, because this would destroy the distinction between forecast and observation.

Verification record

A verification record links a forecast identifier to the matched observation and stores the time of verification, observed value, signed error, absolute error and any quality flags. If the target observation is unavailable, the forecast remains pending rather than being matched to a different timestamp.

If an observation is later corrected, a revised verification record should preserve the reason and time of correction. Monitoring metrics should indicate which observation version was used when material corrections affect reported results.

APPENDIX Q: API AND SERVICE CONTRACT CONSIDERATIONS

Request validation

An inference service should validate the requested variable, issue time, required input fields and feature schema. Invalid requests should return clear errors rather than a numerical value with missing provenance. A successful response should include both the forecast value and its target time.

The service should reject or flag stale inputs according to a configured policy. That policy should be documented and tested, because an old observation may be particularly misleading during rapidly changing conditions.

Response contract

A response can contain a status, issue timestamp, target timestamp, variable name, predicted value, unit, model version and input-data timestamp. RH responses should include the calculation method and identify the predicted temperature and dew point used.

The exact field names and route paths must follow the implemented project API. This appendix defines a recommended contract, not an assertion that these precise names already exist.

Failure handling

Failure states include database unavailable, missing required fields, insufficient history, model artefact missing, invalid numeric output and temperature forecast unavailable for RH. Each state should have a distinct log message and user-facing status.

The service should not fabricate fallback values. If a fallback model is introduced, the result should identify the fallback and record why it was used.

APPENDIX S: MODEL MONITORING METRICS

Table 8-Model monitoring items

Monitoring item	Definition	Why it matters	Suggested review
Data freshness	Age of newest usable observation	Detects stale inputs	Each ingestion/inference cycle
Missingness rate	Share of missing values by field	Detects feed/schema problems	Daily and by source
Verification coverage	Verified forecasts / issued forecasts	Shows completeness of evaluation	Daily/weekly
MAE	Mean absolute error by variable	Tracks typical error	Rolling window and seasonal
RMSE	Root mean squared error by variable	Highlights large misses	Rolling window and seasonal
Bias	Mean signed error	Detects systematic high/low forecasts	Rolling window
Input distribution shift	Change in feature distribution	Potential drift indicator	Periodic
Prediction failure rate	Share of inference attempts that fail	Service reliability	Daily
Model version	Active trained artefact	Supports audit and rollback	Every forecast
Retraining status	Candidate training and acceptance result	Tracks maintenance process	Every retraining event

Interpreting monitoring signals

A single alert rarely identifies the cause of a problem. Increased MAE may arise from unusual weather, a broken input feed, incorrect feature construction or model drift. Investigate data and software health before concluding that the model needs retraining.

Monitoring windows should be long enough to provide a useful sample while remaining responsive to changes. Report the number of verified forecasts with every metric and avoid comparing windows with substantially different coverage without qualification.

APPENDIX U: RH CALCULATION AND VALIDATION WALK-THROUGH

This appendix explains the sequence required to calculate and validate relative humidity. It does not introduce additional empirical results. The input values for a real evaluation must be obtained from the model prediction records.

Required inputs

The calculation requires a predicted dew-point temperature and a predicted dry-bulb temperature for the same forecast target time. Both values must be expressed in degrees Celsius. The target timestamps should match exactly, and each input should have a documented model version or source record.

If either value is missing, non-numeric or not finite, the RH calculation should return an unavailable status. It should not substitute a value from a different target time without a clearly documented policy. This prevents a technically valid formula from producing a misleading forecast.

Calculation sequence

First, validate the input types and units. Second, calculate the Magnus term for dew point, 17.67Td/(Td + 243.5), and the corresponding term for dry-bulb temperature, 17.67T/(T + 243.5). Third, subtract the temperature term from the dew-point term, apply the exponential function, and multiply by 100. Finally, validate that the result is finite and within the expected physical range.

The equation is sensitive to the relationship between T and Td. Under ordinary unsaturated conditions, dew point is not expected to exceed dry-bulb temperature. When model outputs violate that relationship beyond an agreed tolerance, the system should flag the case for investigation. Any correction should be explicitly recorded and must not silently alter the model output.

Verification sequence

After the target time has passed and an observation is available, match the forecast RH to the reference RH for the same timestamp and station. Calculate signed error, absolute error and

squared error. Aggregate the results over a stated evaluation interval and report the number of valid pairs. The reference RH source or calculation method should be stated.

If observed RH is calculated from observed temperature and observed dew point using the same equation, describe it as a formula-derived reference rather than an independent RH sensor measurement. If a source-reported RH value is used instead, document its source, precision and any rounding conventions.

Reporting checklist

☐ State the exact RH formula and constants used.

☐ Confirm temperature and dew-point units are degrees Celsius.

☐ Confirm issue and target timestamps are aligned.

☐ Record the number of missing or rejected input pairs.

☐ Report RH MAE, RMSE, signed bias and sample count.

☐ Explain whether reference RH is directly observed or formula-derived.

☐ Report any out-of-range outputs and how they were handled.

☐ Do not infer RH accuracy from dew-point MAE alone.

Interpretation

The RH calculation is a deterministic transformation of two model outputs. It can improve consistency of the displayed variables, but it cannot remove error already present in temperature or dew point. Therefore, RH is a separate evaluation target even though it is not produced by an independent regression model.

The final thesis should only report completed RH validation after the prediction records and matching observations have been processed. Until then, the correct conclusion is that the method is specified, while its quantitative RH performance remains unreported in the supplied group paper.